

6th Grade Unit 12 Assessment Guide

General Information	The benchmarks on this assessment are closely related yet distinctive in their expectations. For example, MA.6.AR.1.1 has students write expressions whereas MA.6.AR.2.2 and MA.6.AR.2.3 has students write equations. The cognitive demand is similar for both of these benchmarks. Interpreting real-world contexts can be difficult for students. Providing them with a variety of contexts and encouraging students to draw models and/or pictures to help support their thinking process can help students move to the abstract. When solving an equation, students may show work using algebraic steps, models, or pictures. When solving equations students, may write the variable as x instead of given variable. Consider a small deduction of points if a student consistently makes this error throughout the assessment, but this should not be overemphasized at this grade level. Consider providing feedback to students if making this error rather than deducting points. The focus should be on the student's ability to solve the equations. Clarification 1 of benchmark MA.6.AR.2.4 states, "Instruction focuses on using algebraic reasoning, drawings, and mental math to determine unknowns." The Grade 6 B1G-M states, "The benchmark expects students to reason and use mental math to build their number sense. Students should not follow a specific set of steps to solve." Questions 1 and 15 are set up so that the student is using algebraic reasoning or mental math to solve it.
Calculator Information	All questions are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.6.AR.2.4
	The value of h can be determined to be ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ because ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☒ $\frac{37}{7}$ ☐ $\frac{52}{7}$ is ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ ☐ less than ☒ more than ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ OR The value of h can be determined to be ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ because ☒ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ is ☐ $\frac{15}{7}$ ☒ $\frac{22}{7}$ ☐ $\frac{37}{7}$ ☐ $\frac{52}{7}$ ☒ less than ☐ more than ☐ $\frac{15}{7}$ ☐ $\frac{22}{7}$ ☒ $\frac{37}{7}$ ☐ $\frac{52}{7}$	Recommended Scoring (10 Total Points)	Consider deducting 2 points for each incorrect response box.

Question	2	Benchmark(s)	MA.6.AR.1.1
☐ fourteen less than y ☐ y less than fourteen ☐ the sum of y and -14 ☐ y subtracted from -14 ☐ the difference between y and 14		Recommended Scoring (15 Total Points)	Consider giving students 15 points for having the three correct selections. Consider deduction of 5 points for each incorrect selection.
Question	3	Benchmark(s)	MA.6.AR.1.1
Expression $300 - 4h$		Recommended Scoring (5 Total Points)	Students may use a different variable, but it's recommended not to deduct points for this error. Consider giving students 2 points for $300 - h$.
Question	4	Benchmark(s)	MA.6.AR.2.3
$x = -18$		Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $x = 18$ if the student simply lost the negative sign.

Question	5	Benchmark(s)	MA.6.AR.2.2
	$b = 135$	Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points if the student simply lost the negative sign.
Question	6	Benchmark(s)	MA.6.AR.2.2
	$w = -11$	Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $w = 11$ if the student simply lost the negative sign.
Question	7	Benchmark(s)	MA.6.AR.2.3
	$y = 144$	Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $y = -144$ if the student simply lost the negative

Question	8	Benchmark(s)	MA.6.AR.2.3
Equation $50x = 3100$		Recommended Scoring (5 Total Points)	Students may use a different variable, but it's recommended not to deduct points for this error. Students may choose to write the equation as $\frac{3100}{50} = x$ or $\frac{3100}{x} = 50$. Students should not lose points for correct mathematics.
Question	9	Benchmark(s)	MA.6.AR.2.2
$k = -24$		Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $k = 24$ if the student simply lost the negative sign.
Question	10	Benchmark(s)	MA.6.AR.2.3
$h = -225$		Recommended Scoring (5 Total Points)	Students may choose to show their work in a variety of ways. Consider giving students 2 points for $h = 225$ if the student simply lost the negative.

Question	11a	Benchmark(s)	MA.6.AR.2.2
Equation $x - 112 = 204$		Recommended Scoring (5 Total Points)	Students may use a different variable, but it's recommended not to deduct points for this error. Students may choose to write the equation as $x - 204 = 112$ or $x = 204 + 112$. Students should not lose points for correct mathematics.
Question	11b	Benchmark(s)	MA.6.AR.2.2
316 minutes		Recommended Scoring (5 Total Points)	Students may get this correct and miss question 11a.
Question	12	Benchmark(s)	MA.6.AR.2.3
$\frac{c}{75} = 4$		Recommended Scoring (5 Total Points)	Consider giving no partial credit.
Question	13	Benchmark(s)	MA.6.AR.1.1
Expression $\frac{n}{8}$		Recommended Scoring (5 Total Points)	Students may write the answer as $n \div 8$.

Question	14	Benchmark(s)	MA.6.AR.2.2
Equation $x + 36 = 110$		Recommended Scoring (5 Total Points)	Students may choose their variable. Students may write the answer as $x = 110 - 36$ or $110 - x = 36$.
Question	15	Benchmark(s)	MA.6.AR.2.4
The value of h in the bar model is ☐ -2.1. ☐ 5. ☒ -10.5. The bar model represents the equation ☐ $-2.1h = -5$. ☒ $\frac{h}{5} = -2.1$. ☐ $h = -2.1$.		Recommended Scoring (10 Total Points)	Consider giving 5 points for each correctly completed statement.